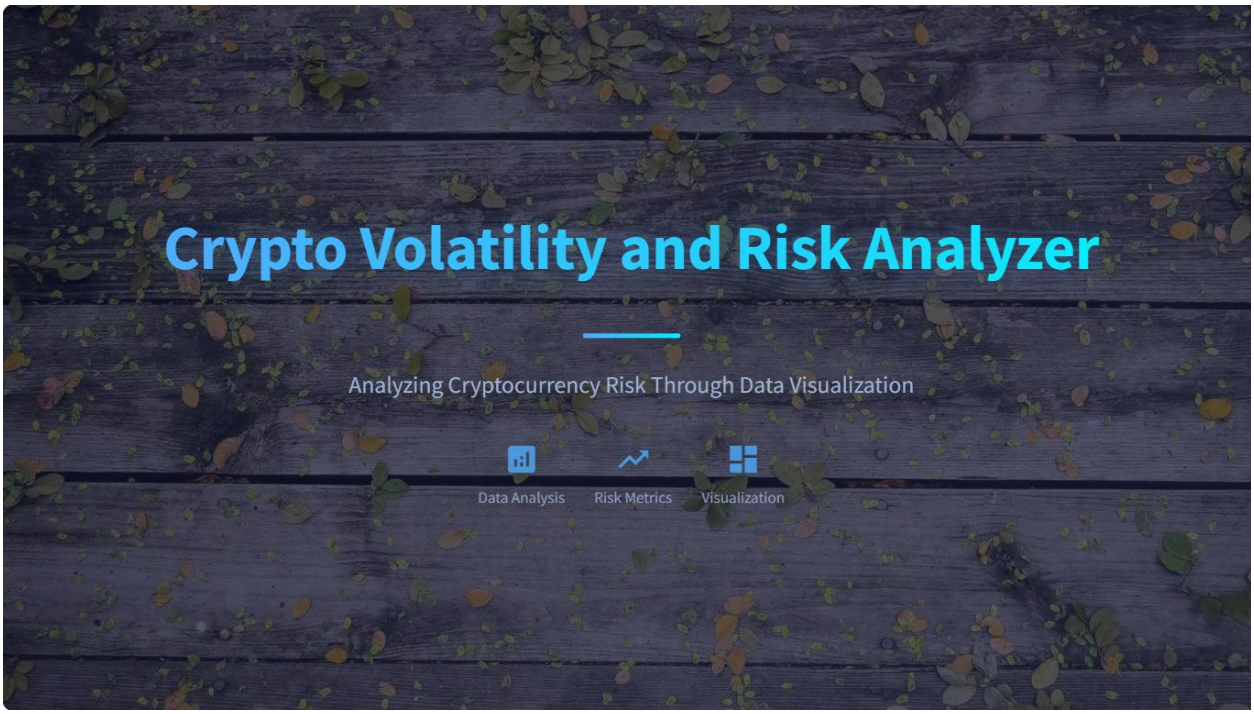
Crypto Volatility and Risk Analyzer



Project Statement

The project **"Crypto Volatility and Risk Analyzer"** aims to analyze and visualize the volatility patterns of selected cryptocurrencies to estimate their risk levels over time. The system fetches real-time and historical cryptocurrency data from APIs such as CoinGecko or Binance and performs quantitative analysis to measure volatility, Sharpe ratio, and beta coefficients.

Using data analytics and visualization libraries in Python (like Matplotlib and Plotly), this project will generate interactive visual dashboards that display risk insights. The tool will ultimately help traders, investors, and researchers identify high-risk and low-risk cryptocurrencies through dynamic and data-driven visual analysis.

Outcomes

1. **Dynamic Data Fetching:**
   Automatically fetch live and historical cryptocurrency data using APIs (Binance / CoinGecko).

2. **Risk Analysis Metrics:**
   Compute financial indicators including daily and annualized volatility, Sharpe ratio, and beta coefficients.
3. **Interactive Visualizations:**
   Build real-time visual dashboards using Plotly or Matplotlib to depict price trends and volatility movement.
4. **Risk Categorization:**
   Classify cryptocurrencies into High-Risk, Medium-Risk, and Low-Risk groups based on computed metrics.
5. **Decision Support:**
   Provide actionable insights for investors by visually comparing volatility and return trade-offs.

Modules to be Implemented

1. **Data Acquisition Module**
   - Fetch live and historical price data for multiple cryptocurrencies.
   - Handle missing values and format data for analysis.
   - Data source integration (CoinGecko, Binance).
2. **Data Processing and Calculation Module**
   - Compute daily returns, volatility, Sharpe ratio, beta coefficient, and Value-at-Risk (VaR).
   - Apply rolling window methods to track time-based variations.
3. **Visualization Module**
   - Create interactive graphs such as line charts, scatter plots, and risk-return maps using Plotly or Matplotlib.
   - Display comparative volatility trends across multiple assets.
4. **Risk Classification and Reporting Module**
   - Categorize cryptocurrencies based on volatility and Sharpe ratio thresholds.
   - Generate summarized reports with key insights and visual exports.

Week-wise Module Implementation and High-Level Requirements

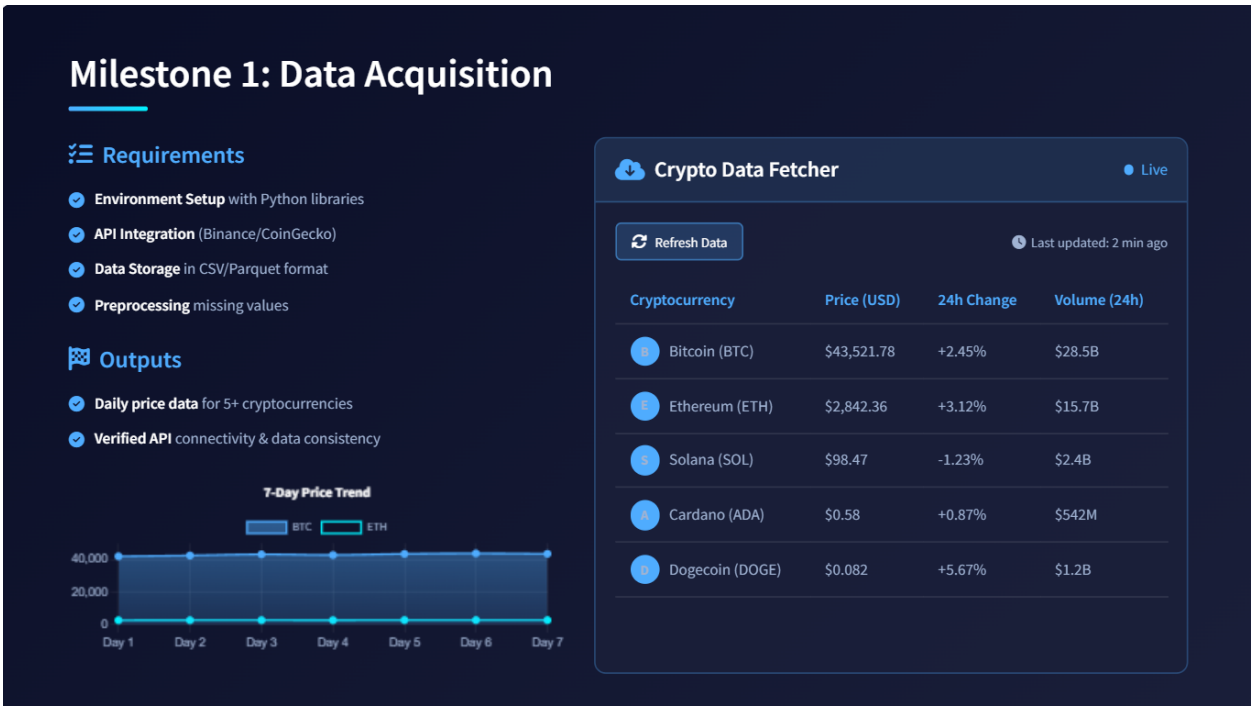
Milestone 1 – Week 1 & Week 2: Data Acquisition and Setup

Requirements:

- Set up project environment and libraries (pandas, numpy, requests, plotly, dash).
- Integrate APIs (Binance/CoinGecko) for fetching crypto price data.
- Store fetched data into local storage (CSV/Parquet format).
- Perform basic preprocessing – missing value handling and time-series formatting.

Output:

- Successfully retrieved and stored daily price data for at least 5 cryptocurrencies (e.g., BTC, ETH, SOL, ADA, DOGE).
- Verified API connectivity and data consistency.



Milestone 2 – Week 3 & Week 4: Data Processing and Calculation

Requirements:

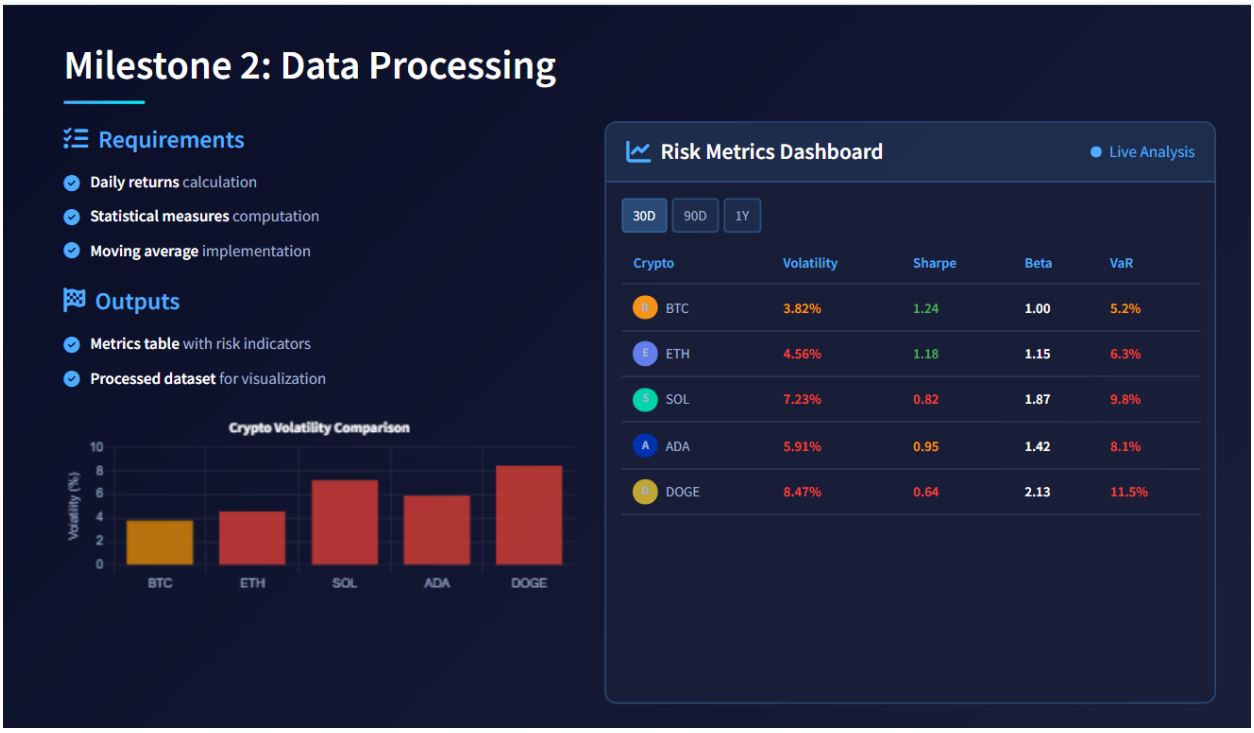
- Calculate daily returns using log-return formulas.
- Derive statistical measures:

- Daily and annualized volatility
- Sharpe ratio
- Beta coefficient (vs benchmark like BTC).

- Implement moving average and rolling window volatility.

Output:

- Generated a metrics table with volatility, Sharpe ratio, and beta values for all selected assets.
- Prepared intermediate dataset for visualization and risk classification.



Milestone 3 – Week 5 & Week 6: Visualization and Dashboard Development

Requirements:

- Build interactive visualizations using Plotly or Matplotlib.
- Develop an analytical dashboard (Streamlit/Dash) to:
  - Display time-series graphs for price and volatility.
  - Show risk-return scatter plots.
  - Enable multi-crypto comparisons and filtering.

Output:

- Interactive dashboard visualizing volatility and Sharpe ratio for multiple cryptocurrencies.
- Responsive UI enabling date-range selection and asset comparison.

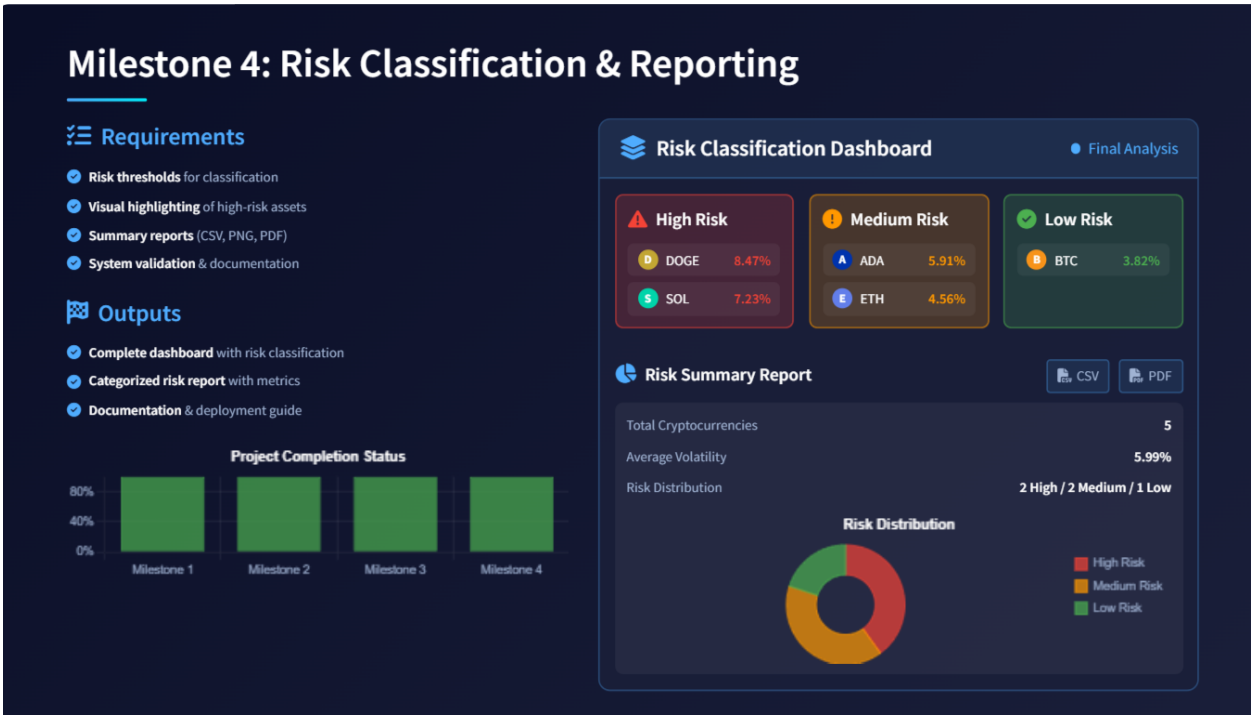
Milestone 4 – Week 7 & Week 8: Risk Classification and Reporting

Requirements:

- Define thresholds for classifying risk levels (Low, Medium, High).
- Highlight high-risk assets visually in the dashboard.
- Generate final summary reports (CSV, PNG, or PDF export).
- Validate system performance and finalize documentation.

Output:

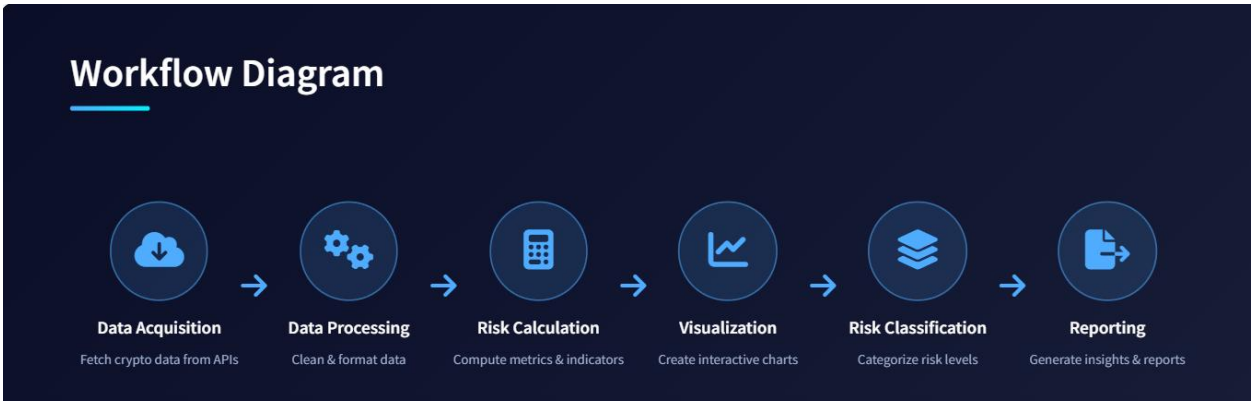
- Complete “Crypto Volatility and Risk Analyzer” dashboard.
- Final categorized risk report summarizing volatility levels and Sharpe ratios for each crypto asset.
- Documentation and user guide prepared for deployment.



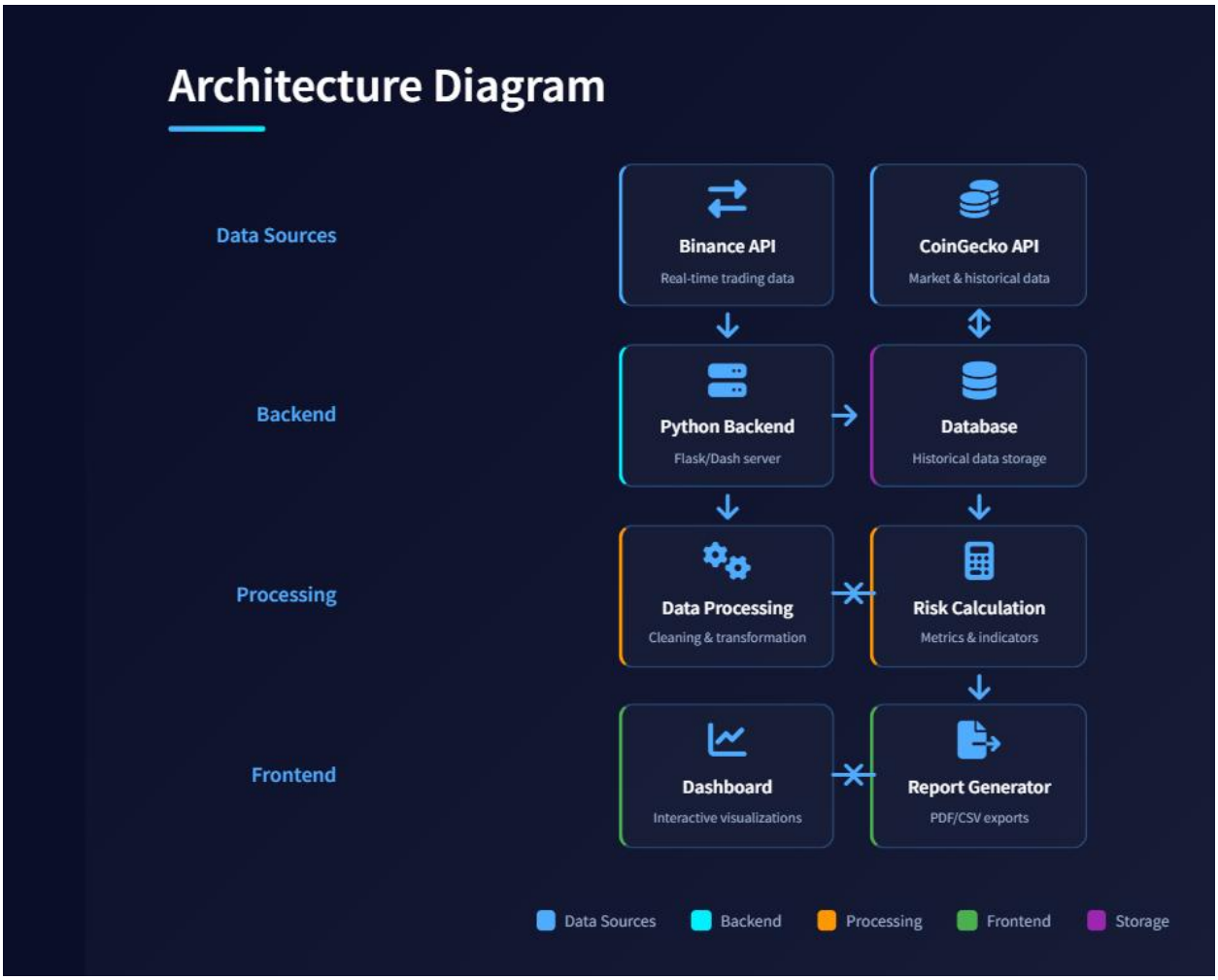
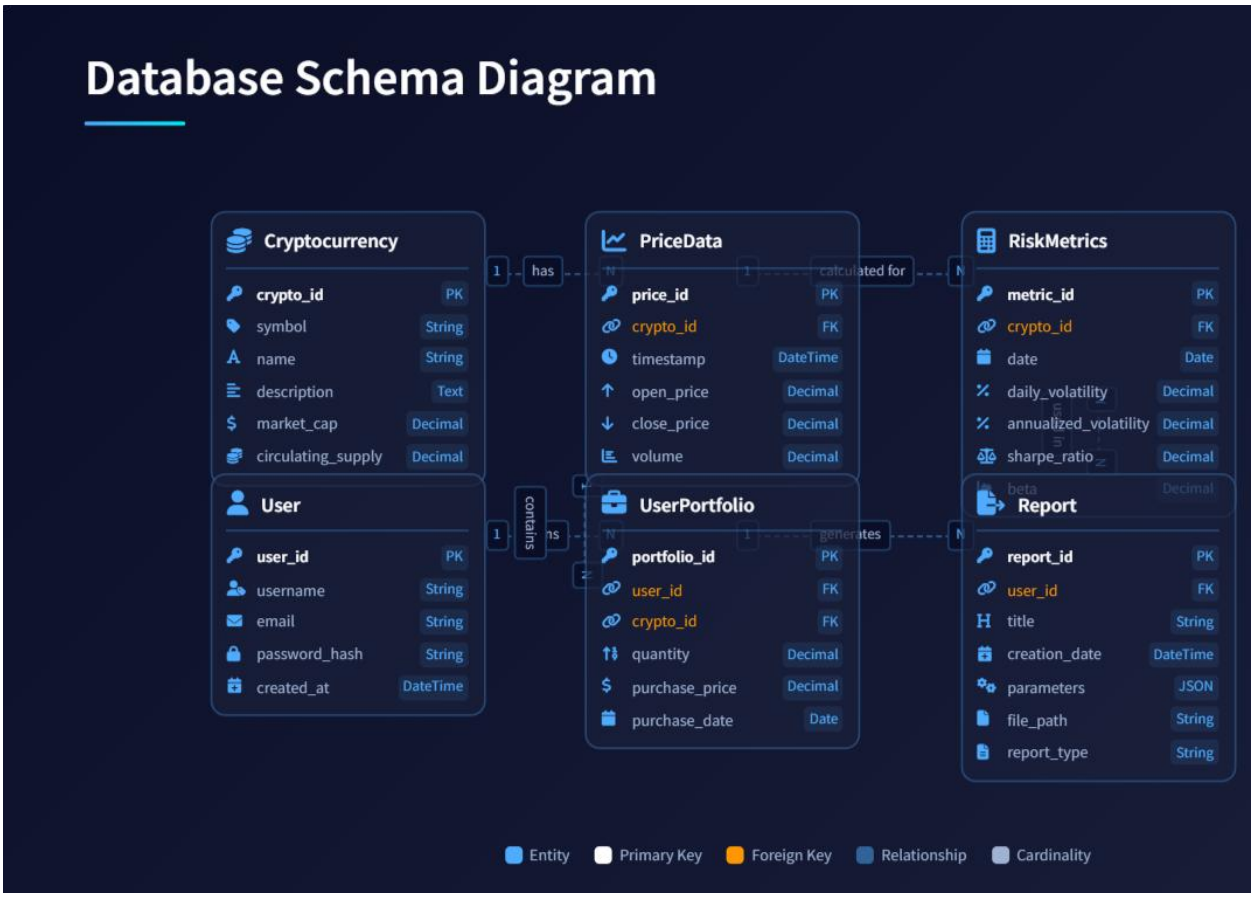
Evaluation Criteria

Parameter	Description
Milestone 1	Successful API data fetching, preprocessing, and storage.
Milestone 2	Correct computation of daily volatility, Sharpe ratio, and beta coefficient.
Milestone 3	Creation of interactive and meaningful visualizations using Plotly or Matplotlib.
Milestone 4	Implementation of accurate risk classification and report generation.
Documentation & Presentation	Clear explanation, well-structured code, and user-friendly dashboard interface.

Work Flow Diagram



Architecture Diagram



Schema Diagram